

Avanindraa Nagarajan

Computer Science Engineering Student (Specialization in Artificial Intelligence)

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A reliable and motivated third-year student with a strong foundation in computer science. A leader having strong written and verbal communication skills. Proficient in problem-solving and open to learning new technologies.

A go-getter who is passionate about software development with a strong interest in building efficient and scalable solutions using AI agents and tools. Enthusiastic about collaborating with teams and implementing innovative features to create impactful technology solutions.

Skills

Strategic Planning & Execution | Programming Languages | Problem-Solving | Collaborative Development | Cross-Functional Teamwork | Effective Communication |

Education

COLLEGE

Amrita Vishwa Vidyapeetham Chennai Campus, Tamil Nadu, India

- Third year engineering student (Batch: 2023-2027)

SCHOOLING

APL Global School, Tamil Nadu, India

- Completed 'A' Level (GCE)
- Completed IGCSE (Cambridge)

Certifications

- Database Management System (DBMS) & SQL: Complete Pack (Infosys)
- Essentials of Cloud Computing (Infosys)
- Theory of Computation (NPTEL)

Projects

Enhancing Fokker-Planck Equation Solutions with Physics-Informed Neural Networks:

- Conducted a research project exploring numerical solutions of the Fokker-Planck equation using Physics-Informed Neural Networks.
- Integrated Jensen-Shannon and Kullback-Leibler Divergence measures to improve accuracy in solving differential equations.
- Developed and tested models using Python and MATLAB, gaining practical experience in computational modelling and data analysis.

Probabilistic Error Detection in Large Language Models:

- Designed an uncertainty-aware fact-checking framework using Bayesian Networks to detect factual errors in LLM-generated claims.
- Integrated transformer-based classifiers with an external NLI fact-checker to model semantic agreement and confidence signals.
- Performed probabilistic inference by combining model confidences, agreement indicators, and contextual metadata.
- Evaluated the system on benchmark datasets and achieved improved accuracy, calibration, and interpretability in error detection.

Cross-Modal Video Summarization with Emotion-Trajectory and Audio-Semantic

Alignment:

- Designed a multimodal video summarization framework integrating visual, audio-emotion, and audio-semantic signals for enhanced summary generation.
- Extracted CLIP-based visual embeddings and aligned them with audio-emotion trajectories using pretrained models (OpenL3, VGGish, YAMNet, CREMA-D).
- Developed an audio-visual temporal alignment mechanism to detect emotion peaks and semantically significant acoustic events.
- Engineered temporal prompt tokens and audio-aware highlight scoring to improve contextual importance modeling.
- Integrated multimodal embeddings into LLaMA/Mistral-based language decoders for video-to-text and video-to-video summary generation.

Technical Expertise

- Web Designing and development (HTML, CSS and JavaScript)
- Microsoft products: Word, Excel, PowerPoint
- Programming Languages:
 - Python
 - C
 - Java
- MATLAB
- AI agents and tools

Achievements

- Participated in the Adobe India Hackathon 2025 and reached Quarter-finals (Built an intelligent PDF agent)
- Participated in Smart India Hackathon (SIH 2023 and 2024)
- Published "Classification of White Blood Cells using Feature Extraction and Cell Segmentation" at the IEEE International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT) 2025.